

Gagandeep Singh

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Skills

Programming: Go, Python, Rust, Java, Scala

System Architecture & Backend: Layered Architecture, Event-Driven Architecture, Domain-Driven Design, Microservices, Distributed Systems, Test Driven Development, REST (HTTP), gRPC

AI & MLOps: Argo, Airflow, MLflow, Langchain, RAG, AI Agents

Data Science & Machine Learning: TensorFlow, PyTorch, Hugging Face, Scikit-Learn, Pandas, Numpy, SHAP

Data Management: Postgres, DynamoDB, Redis, Kafka, Cassandra, Qdrant

Cloud Computing: AWS, Azure, Docker, Kubernetes, Datadog, Terraform, CircleCI

Experience

Deliveroo (acquired by DoorDash)

Feb. 2022 – Present

Senior Software Engineer – White Label Delivery as a Service

Mar. 2026 – Present

- Developed and enhanced Deliveroo Express, a white label DaaS platform processing 900,000 deliveries and contributing £10.4M+ in annualised revenue via an event-driven Go/Rust polyglot architecture
- Designed and led delivery of an external order tracking system by designing a secure token-based real-time tracking solution that unblocked commercial contracts with enterprise retail partners

Senior Software Engineer – Ad Platform, Financial Systems & MLOps

Apr. 2025 – Feb. 2026

- Developed and enhanced the Ad Platform managing 45,000+ campaigns, contributing £100M+ annual revenue
- Scaled ML serving platform to 1000 RPS driven by restaurant network growth and multi-campaign support, resolving p99 latency regression from 30ms to sub-20ms through inference batching optimisations
- Designed and implemented Targeting and Scheduling for the Ad Platform, generating £5M+ in revenue boosting advertiser ROI, engagement and campaign flexibility
- Redesigned and led a phased migration to decouple the Ad Platform from the financial system, introducing asynchronous communication to remove hard dependencies, streamline team workflows and accelerate feature delivery

Software Engineer – Ad Platform, Financial Systems & MLOps

Feb. 2022 – Mar. 2025

- Co-led rollout of ads in filtered lists and offers, boosting ad visibility and driving a £2.5M uplift in annual revenue
- Engineered a low-latency ML serving platform for click probability prediction, auto-bidding, relevance models achieving sub-20ms p99 latency with end-to-end MLOps lifecycle and A/B experimentation support
- Unlocked £3.5M annual revenue by delivering the Ad Credits system & Co-Funded Campaigns for partner incentives
- Engineered a Centralised Billing System supporting complex franchise hierarchies and cross-partner campaigns, automating invoicing workflows and unlocking £1.92M in operational efficiency
- Built a proof-of-concept LLM-powered AI agent for campaign management leveraging tool calling

Thoucentric

Nov. 2020 – Jan. 2022

Machine Learning Engineer – FMCG Products

Bangalore, India

- Developed robust forecasting and commodity pricing models, improving prediction accuracy and reducing training times by 45% through containerisation and distributed processing
- Spearheaded design and development of scalable ML pipelines with distributed task execution across Kubernetes clusters, optimising concurrency and resource usage
- Designed and implemented scalable hyper-parameter tuning and model lifecycle frameworks, improving automation and reducing development time by 30%
- Built causal analysis and visualisation notebooks enabling teams to explore scenarios and make data-driven decisions

Personal Projects

J.A.M.S – Just Another Model Server | [Link](#)

Rust, OpenTelemetry, Seldon, Cloud-Agnostic

- Built a cloud-agnostic low latency, high throughput model server with multi-framework support and observability

Real-Time Credit Card Fraud Detection | [Link](#)

Python, Spark, Streaming ML, ETL, Orchestration

- Implemented a streaming anomaly detection system for credit card transactions using Spark ML

PaSys – Payment & Settlement System | [Link](#)

Rust, Java, Spring Boot, Event-Driven, ACID Ledger

- Designed an event-driven fault tolerant system with ACID guarantees, reconciliation and multi-party settlements

Education

University Of Nottingham

Nottingham, United Kingdom

MS Data Science

Sep. 2019 – Sep. 2020